Assignment 3

Econ 8007 Macroeconomic Theory I, Fall 2026

Due on Friday, October 16th, 2026

Part A: Theory

A1. A household lives two periods and solves

$$\max_{c_1,c_2,h_1,h_2,s} \ \ln c_1 - B\,\frac{h_1^{1+1/\nu}}{1+1/\nu} + \beta\Big[\ln c_2 - B\,\frac{h_2^{1+1/\nu}}{1+1/\nu}\Big]$$

subject to $c_1 + s = a_0 + w_1 h_1$, $c_2 = (1+r)s + w_2 h_2$ and $s \geq -\bar{a}$. Let $R \equiv a_0 + w_1 h_1 + w_2 h_2/(1+r)$ denote lifetime resources and $\phi_1 \equiv w_1 h_1/R$, $\phi_2 \equiv w_2 h_2/[(1+r)R]$ the shares of the two labor incomes in R.

(a) Write the first-order conditions with multipliers λ_1, λ_2 on the budget constraints. Show that the Frisch elasticity of h_1 with respect to w_1 (holding λ_1 fixed) equals ν, and explain in one sentence why holding λ_1 fixed is the right thought experiment for a business-cycle model but not for a study of permanent wage differences across workers.

(b) *Permanent change.* Assume the borrowing constraint is slack. Let w_1 and w_2 rise by the same proportion. Show that h_1 and h_2 do not change when $a_0 = 0$, and sign the response when $a_0 > 0$. Which two effects cancel, and why does the cancellation depend on non-labor wealth?

(c) *Temporary change.* Still unconstrained, let only w_1 rise. Show that

$$\varepsilon^{\text{temp}} \equiv \frac{d\ln h_1}{d\ln w_1} = \nu\,\frac{1-\phi_1+\nu\phi_2}{1+\nu\phi_1+\nu\phi_2},$$

and verify that it lies between 0 and ν. When is it close to ν, and when is it close to zero? Relate the two limits to (a) and (b).

(d) *A constrained household.* Now suppose the constraint binds, $s = -\bar{a}$, and $a_0 + \bar{a} = 0$. Show that h_1 does not respond to w_1 at all. So the constrained household has a *smaller* wage elasticity than the unconstrained one. Reconcile this with the claim, made in class, that hours are an insurance margin that is most valuable exactly where the asset margin is weakest—what is it that the constrained household's hours respond to, if not the wage? (Hint: write h_1 as a function of a period-1 non-labor income shock.)

(e) *Precautionary labor supply.* Return to the unconstrained household and let period-2 non-labor income be $\tilde{y}$ with $E\tilde{y} = 0$ and variance σ^2, so $c_2 = (1+r)s + w_2 h_2 + \tilde{y}$. Show that a small increase in σ^2 raises h_1, and identify which condition (the Euler equation or the intratemporal condition) carries the effect. Why is this channel absent for the household in (d), even though that household is the one most exposed to the risk?

A2. Consider the following one-period economy. Each household enters with an asset endowment $a \geq 0$ (a stock it already holds, distributed across households with cdf G) and a productivity draw e, uniform on $[0,1]$ and independent of a. There is no intertemporal decision: the household liquidates its assets, which pay $(1+r)a$, and consumes all of its resources. Its only choice is whether to work $\bar{h}$ hours, at a fixed utility cost $\bar{B}$, or not at all. Utility is $\ln c - \bar{B}\,\mathbf{1}\{\text{work}\}$, so a household that works consumes $(1+r)a + we\bar{h}$ and one that does not consumes $(1+r)a$. Set $\bar{B} = 0.3$, $\bar{h} = 1/3$, $w = 1$, $r = 0.04$. (Think of a as the wealth a household has accumulated before the period starts; the point of the exercise is how the distribution of that wealth shapes the aggregate response, which the dynamic model of B2 then endogenizes.)

(a) Derive the reservation productivity $e^*(a) = \kappa a$ and the number κ. Which households work regardless of e, and which never work?

(b) Show that aggregate employment is $N(w) = \int \max\{0,\ 1 - e^*(a;w)\}\, dG(a)$ and derive the aggregate employment elasticity

$$\varepsilon_N \equiv \frac{d\ln N}{d\ln w} = \frac{1}{N}\int \mathbf{1}\{e^*(a;w) < 1\}\; e^*(a;w)\, dG(a).$$

Explain why only the households with $e^*(a) < 1$ appear in the numerator.

(c) Compute N and ε_N for three wealth distributions with the same mean, $Ea = 0.5$: (i) everyone holds $a = 0.5$; (ii) half hold $a = 0.05$ and half hold $a = 0.95$; (iii) half hold $a = 0$ and half hold $a = 1$. Comment on the range of elasticities you find, and on who is at the margin in each economy.

(d) In economy (ii), the government transfers 0.05 units of wealth from each rich household to each poor household. Does the elasticity rise or fall? Why does this matter for evaluating a fiscal experiment with the "aggregate labor supply elasticity" one has calibrated beforehand?

(e) Contrast your numbers with two benchmarks: the intensive-margin Frisch elasticity ν of a single worker, and the aggregate elasticity of the Hansen–Rogerson lottery economy. In a paragraph, explain how the same preferences can generate a micro elasticity of 0.5, an aggregate elasticity anywhere between 0 and above 1, and an infinite elasticity, and what the smoothing mechanism is in each case.

A3. A worker lives for T periods, $t = 0, \ldots, T-1$, has human capital h_t rented at the price w, and splits her time between earning and investing: a fraction $s_t \in [0,1]$ is invested, so period earnings are $wh_t(1-s_t)$, and human capital accumulates according to

$$h_{t+1} = (1-\delta)h_t + a\,(s_t h_t)^{\alpha}, \qquad \alpha \in (0,1),$$

where a is learning ability. There is no uncertainty and utility is linear, so the worker chooses $\{s_t\}$ to maximize $\sum_{t=0}^{T-1} \beta^t w h_t (1 - s_t)$. Write $x_t \equiv s_t h_t$ for the investment input and $Q_t \equiv V'_{t+1}(\cdot)$ for the marginal value of a unit of human capital carried into $t+1$, where $V_t(h)$ is the value function. Recall from class that the envelope condition gives $Q_{t-1} = w + \beta(1-\delta) Q_t$ with $Q_{T-1} = 0$, so that

$$Q_t = w \, \frac{1 - [\beta(1-\delta)]^{T-1-t}}{1 - \beta(1-\delta)}, \qquad x_t = \left(a \alpha \beta Q_t\right)^{1/(1-\alpha)} \ \text{(with } w = 1\text{)}$$

at an interior optimum. Set $w = 1$ unless stated otherwise.

(a) *Two periods.* Let $T = 2$ (periods 0 and 1; no investment at $t = 1$). Solve for the optimal investment input $x_0 = s_0 h_0$ and show it does not depend on h_0. When does the corner $s_0 = 1$ bind? Interpret the corner.

(b) *Fanning.* Still with $T = 2$, consider two workers with the same h_0 and abilities $a_H > a_L$, both interior. Show that the high-ability worker has lower earnings at $t = 0$ and higher earnings at $t = 1$, so the profiles cross. Compute the individual growth rate $g_i = \ln y_{i1} - \ln y_{i0}$ and show that it is increasing in a_i. What would an econometrician who fits $y_{it} = \alpha_i + \beta_i t$ to a large cross-section of these workers conclude about the source of the rising variance of earnings? Is that conclusion right?

(c) *General T.* Using the closed forms for Q_t and x_t above, sign $\partial x_t / \partial T$ and $\partial x_t / \partial \delta$ and interpret: what happens to investment at every age if the retirement age is raised by five years?

(d) *The price of skill.* Show that a permanent proportional change in w leaves the whole path $\{s_t\}$ unchanged. Now suppose the skill price w grows at a constant rate g_w over the worker's life, unbeknownst to the econometrician. Explain why the measured earnings profile confounds the price of skill with the quantity $h_t(1 - s_t)$, and what this implies for using earnings growth alone to estimate a.

A4. In U.S. administrative data, one-year changes in log earnings have a standard deviation of roughly 0.35–0.5 and a kurtosis of about 15, against 3 for a Gaussian. A simple process that reproduces this is a mixture: with probability 0.9, $\eta \sim N(0, 0.20^2)$; with probability 0.1, $\eta \sim N(0, 0.92^2)$. Consider a two-period household with CARA utility $u(c) = -e^{-\gamma c}/\gamma$, $\beta(1+r) = 1$, income 1 in period 1 and $1 + \eta$ in period 2, saving s at the gross rate $1 + r$ (set $r = 0$ for the numbers), and no borrowing constraint.

(a) Verify that the mixture (M) has variance $\sigma^2 \approx 0.12$ and kurtosis $3\,[0.9\,\sigma_1^4 + 0.1\,\sigma_2^4]/\sigma^4 \approx 15$. Let (G) denote a Gaussian $N(0, \sigma^2)$ with the same variance.

(b) Show that the Euler equation delivers

$$s = \frac{1}{2\gamma} \, \ln E\, e^{-\gamma \eta},$$

so that $s_G = \gamma \sigma^2 / 4$ under (G) and $s_M = \frac{1}{2\gamma} \ln \left[0.9\, e^{\gamma^2 \sigma_1^2/2} + 0.1\, e^{\gamma^2 \sigma_2^2/2}\right]$ under (M). Prove $s_M > s_G$ using the convexity of $e^{\gamma^2 x/2}$ in x, and interpret: what feature of the shock distribution, beyond its variance, does the precautionary motive respond to?

(c) Compute s_G and s_M for $\gamma = 2$. Compute also the risk premium $\pi = \frac{1}{\gamma} \ln E\, e^{-\gamma \eta}$ (the certain income loss the household would accept to remove the risk) under each distribution. By what factor does a model calibrated to the "right" variance but Gaussian shocks understate the welfare cost of risk?

(d) Now impose a borrowing constraint $s \geq 0$ and suppose that instead of a smooth optimum the household simply targets $c_2 \geq 0.5$ in the worst realistic realization (a 2-standard-deviation draw from the component that occurs). How large a buffer does that require under (G) and under (M)? Discuss why a household can self-insure against a shock of 0.20 but not against one of 0.92, and what the large draws correspond to in the data (think of displacement and long unemployment spells).

A5. In Krusell, Mukoyama and Şahin (2010), households with CRRA utility self-insure with capital against unemployment risk generated by a DMP labor market, and the wage is bargained between a risk-neutral firm and a risk-averse worker whose outside option depends on her wealth. Strip the problem to a static bargain: a worker with cash on hand a consumes $a + w$ if the bargain succeeds and $a + b$ if it fails; the firm gets $z - w$ or zero. With worker utility u and bargaining weight γ the wage solves

$$\max_{w} \; \big[u(a+w) - u(a+b)\big]^{\gamma} (z-w)^{1-\gamma}.$$

(a) Write the first-order condition. Show that with a risk-neutral worker it gives $w^{RN} = \gamma z + (1-\gamma) b$, and that with any strictly concave u the bargained wage is *below* w^{RN}. (Hint: $u(a+w) - u(a+b) > u'(a+w)(w-b)$.)

(b) Take $u = \ln$. Show that w is increasing in a and converges to w^{RN} as $a \to \infty$. Explain the economics in words: what exactly about being poor weakens the worker's position in the bargain? Is it that the poor worker is "more risk averse"?

(c) Now take $u(c) = -e^{-\theta c}$. Show that the bargained wage does not depend on a at all. What property of preferences, then, generates wealth-dependent wages, and what does this tell you about which utility functions one should use to think about the insurance role of wages?

(d) With $u = \ln$, $z = 1$, $b = 0.4$, $\gamma = 0.5$, compute the wage for $a = 0$ and $a = 10$ (wealth in units of the monthly wage) and compare with w^{RN}. Then raise b to 0.5 and recompute. Does a more generous UI benefit raise the wage more for the poor or for the rich worker, in dollars and in proportion to the initial gap? Interpret.

(e) The results say firms earn more from matching with poor workers. In a DMP economy with free entry, suppose firms could direct their vacancies at workers of a given wealth. Which segment of the wealth distribution would have the higher job-finding rate, and how would that qualify the statement, made in class, that for the wealth-poor the job-finding hazard is nearly the whole insurance technology? What assumption in Krusell, Mukoyama and Şahin rules this out, and why is the wealth dependence of wages quantitatively modest in their calibration? (Think about the size of $w - b$ relative to typical wealth in (d).)

A6. The Ben-Porath model of A3 assumes linear utility, so the worker maximizes discounted earnings and consumption smoothing plays no role. Restore a consumption motive in

the two-period version of A3(a): $u(c_0) + u(c_1)$ with $u = \ln$, no discounting, $r = 0$,

$$c_0 = h_0(1 - s_0) + q, \qquad c_1 = h_1 - q, \qquad h_1 = (1 - \delta)h_0 + a\,(s_0 h_0)^{\alpha},$$

where q is borrowing at $t = 0$, subject to $q \le \bar{q}$.

(a) *Separation.* Suppose $\bar{q}$ is large enough that the constraint is slack. Show that the optimal investment input $x_0 = s_0 h_0$ is the same as with linear utility, $x_0 = (a\alpha)^{1/(1-\alpha)}$, and independent of h_0; consumption is then smoothed by borrowing. State in one sentence the general principle at work and why it requires a perfect credit market.

(b) *Constrained.* Set $\bar{q} = 0$ and assume the constraint binds. Derive the first-order condition for s_0 and show that investment is lower than in (a), with the shortfall governed by the Euler-equation wedge $u'(c_0)/u'(c_1) > 1$. Explain why a worker who cannot borrow treats a unit of study time as more expensive than the forgone wage.

(c) Show that a lump-sum transfer at $t = 0$ raises investment for the constrained worker and leaves it unchanged for the unconstrained one. Then show that, among constrained workers, $\partial x_0/\partial a$ is smaller than in (a). What does this imply for the cross-sectional relation between ability and earnings growth in a population where credit constraints are common—will the fanning of profiles over- or understate ability heterogeneity? How does the answer bear on an HVY-style attribution of lifetime inequality to "initial conditions"?

(d) *Schooling.* In the linear-utility model, full-time schooling ($s_0 = 1$, zero earnings) arises as a corner whenever Q_0 is large. Show that with $u = \ln$ and $\bar{q} = 0$ the corner $s_0 = 1$ can never be optimal. What, then, must be true of any worker observed in full-time schooling? Relate to the roles of family transfers and student loans, and to the claim that in the linear-utility model schooling emerges endogenously, without any schooling technology being assumed.

Part B: Computation

Parameters for B2 and B3 are collected in Table 1; B1 states its own.

Parameter	B2 (annual)	B3 (monthly)
Discount factor β	0.94	0.996
Interest rate r	0.04	0.003
Wage w	1	1
Utility	$\ln c - B h^{1+1/\nu}/(1+1/\nu)$, $\nu = 0.5$	$c^{1-\mu}/(1-\mu)$, $\mu = 2$
Idiosyncratic state	AR(1) in $\ln e$: $\rho = 0.95$, $\sigma_{\ln e} = 0.4$, 7 states (Rouwenhorst)	employment chain with $\lambda_w = 0.45$, $\sigma = 0.02$; income w if employed, $b = 0.4$ if not
Borrowing limit	$a \ge 0$	$a \ge 0$

Table 1: Parameters for Part B.

B1. Take the model of A3 with $T = 40$, $\beta = 0.96$, $\alpha = 0.7$, $\delta = 0.04$, $h_0 = 1$, $w = 1$, and map $t = 0$ to age 25. Whenever the unconstrained input exceeds the stock, $x_t \ge h_t$, the corner $s_t = 1$ binds (full-time investment, zero earnings).

(a) Implement the closed-form solution for $a = 0.15$ and report the share s_t and earnings $h_t(1 - s_t)$ for $t \in \{0, 5, 10, 15, 20, 25, 30, 39\}$. Check that you find seven years of full-time investment at the start of life, a peak in earnings at $t = 31$, and $s_{20} \approx 0.33$; verify one date against a brute-force numerical solution of the Bellman equation.

(b) Now let ability be heterogeneous: a_i equally spaced on $\{0.12, 0.13, \ldots, 0.18\}$, one thousand workers per value, no shocks. Plot the cross-sectional variance of log earnings by age (ignore years in which earnings are zero) and report the increase between ages 30 and 55. Compare with the empirical finding that the cross-sectional variance of log earnings rises by about 0.25–0.40 between ages 25 and 55.

(c) Treat your simulated panel as data. For ages 33–60, compute $\operatorname{var}(\Delta y)$, $\operatorname{cov}(\Delta y_t, \Delta y_{t-1})$ and $\operatorname{cov}(\Delta y_t, \Delta y_{t-j})$ for $j = 2, \ldots, 5$, where y is log earnings with age effects removed. Apply the random-walk-plus-noise identification formulas, $\operatorname{var}(\Delta y) = \sigma_\eta^2 + 2\sigma_\varepsilon^2$ and $\operatorname{cov}(\Delta y_t, \Delta y_{t-1}) = -\sigma_\varepsilon^2$ (for $y_{it} = \alpha_i + z_{it} + \varepsilon_{it}$ with $z_{it} = z_{i,t-1} + \eta_{it}$), to obtain "$\sigma_\varepsilon$" and "$\sigma_\eta$". What would a RIP econometrician conclude, and which moment gives the game away?

(d) Add an i.i.d. transitory shock $\varepsilon_{it} \sim N(0, 0.134^2)$ to log earnings and redo (c). Does the transitory component get recovered? Does the misspecification of the persistent component become easier or harder to detect? Two paragraphs.

B2. Solve the partial-equilibrium household problem with the parameters of Table 1: a household with state (a, e) chooses (c, h, a') to solve

$$V(a, e) = \max_{c, h, a' \geq 0} \ln c - B \frac{h^{1+1/\nu}}{1 + 1/\nu} + \beta \, E\big[V(a', e') \,\big|\, e\big] \quad \text{s.t.} \quad c + a' = (1 + r)a + weh,$$

with $\ln e$ following the discretized AR(1) in the table; prices (r, w) are fixed throughout. Note that the intratemporal condition $Bh^{1/\nu} = we/c$ pins down h given (a, e, a'). Choose B so that average hours in the stationary distribution equal $1/3$.

(a) Report the stationary distribution's mean assets and the hours policy $h(a, e)$ at the median productivity state for $a \in \{0, 1, 2, 5, 10\}$. Compute the correlation of hours and wealth conditional on e (average across states). Interpret through the intratemporal condition: why do hours fall with wealth, and in what sense is that an insurance margin?

(b) *Micro elasticity.* Confirm numerically that the Frisch elasticity of the hours policy is ν: perturb w by $+2\%$ holding the marginal utility of consumption fixed and compare.

(c) *Macro elasticity, divisible labor.* Raise w permanently by 2%, holding r fixed, re-solve, and compute the percentage change in aggregate effective labor $L = \int e \, h(a, e) \, d\lambda^*$ in the new stationary distribution. Report the implied elasticity and explain why it differs from ν (which way does the wealth effect push, and through which households?).

(d) *Macro elasticity, indivisible labor.* Replace the hours choice by $h \in \{0, \bar{h}\}$ with $\bar{h} = 1/3$ and utility cost $\bar{B} = B\bar{h}^{1+1/\nu}/(1 + 1/\nu)$. Recover the reservation productivity $e^*(a)$ from your policy and plot it. Repeat (c) for aggregate employment. Then, in a paragraph, reconcile the three numbers—ν, the divisible-labor aggregate elasticity, and

the indivisible-labor aggregate elasticity—by locating where the mass of households near the threshold sits in the (a, e) distribution.

B3. Solve the following partial-equilibrium problem with the monthly parameters of Table 1. A household is either employed or unemployed; an employed household loses its job with probability σ each month and an unemployed one finds a job with probability λ_w (roughly the empirical monthly separation and job-finding probabilities in the U.S.). Income is $w = 1$ when employed and $b = 0.4$ when unemployed, and the household saves in one riskless asset with $a \geq 0$:

$$V^E(a) = \max_{c,\, a' \geq 0} \; u(c) + \beta\big[(1-\sigma)V^E(a') + \sigma V^U(a')\big], \qquad c + a' = (1+r)a + w,$$
$$V^U(a) = \max_{c,\, a' \geq 0} \; u(c) + \beta\big[\lambda_w V^E(a') + (1-\lambda_w)V^U(a')\big], \qquad c + a' = (1+r)a + b.$$

(a) Report the consumption and saving policies for employed and unemployed households at $a \in \{0, 1, 5, 20\}$ (in months of wages) and the stationary wealth distribution's mean and median. How many months of wages does a household hold, on average, and how does that compare with the expected duration of unemployment $1/\lambda_w$?

(b) *Welfare cost of unemployment risk.* Compare each household with a counterpart facing no risk who receives the ergodic mean income $\bar{y} = (\lambda_w w + \sigma b)/(\lambda_w + \sigma)$ with certainty every month. Compute the consumption-equivalent welfare cost of risk as a function of initial wealth, separately for initially employed and initially unemployed households, and plot it. Where is the cost concentrated?

(c) *Willingness to pay for insurance.* Raise b from 0.4 to 0.5 and compute, again by initial wealth and employment status, the permanent consumption households would give up to live in the high-b economy. (This is a partial-equilibrium number: λ_w and w are held fixed.)

(d) Two paragraphs. First: whom does the Hagedorn–Manovskii calibration, in which the unemployed are nearly indifferent between working and not, describe well, and whom does it describe badly, according to your figures? Second: in a DMP economy with free entry, a higher b shrinks the match surplus, lowers vacancy creation and therefore lowers the job-finding probability λ_w. How would these general-equilibrium effects qualify the willingness to pay you found in (c), and for which households would the qualification bite hardest?